

PRODUCT OVERVIEW

An intelligence layer for essential field services

What PeakLogic is, what it does, and — stated separately — what is built, what is designed, and what is roadmapped.

Document

Artifact #40 · v1.0

Prepared

2 August 2026

Platform

Azure-native · multi-tenant

Status

Pre-deployment

Contents

01	What PeakLogic is	The gap between an autodialer and SCADA
02	The four-step chain	Monitor, predict, dispatch, prove
03	Platform architecture	Five layers and four surfaces
04	Tenant isolation	Why a refusal beats an approximation
05	Product packs	Packaged by what the site actually is
06	Capability by build state	Built, designed, roadmapped
07	Deployment	Azure-native, outbound-only
08	What PeakLogic will not do	Boundaries held on purpose

1. What PeakLogic is

PeakLogic connects the equipment a business already operates and services — pumps, treatment units, chemistry, refrigeration — to monitoring, alerts, automation, and dispatch. Issues surface early, and the right work lands in the right system.

It sits between two bad options. **Autodialers** are cheap and call once the water is already too high. **Traditional SCADA** produces excellent data and is priced and scoped as a capital project. Most distributed assets — lift stations, aerobic treatment units, package plants, commercial pools, walk-in refrigeration, light commercial mechanical rooms — fall into the gap and are monitored by nobody.

Where a plant already runs SCADA, PeakLogic complements it: instrumenting the long tail that platform cannot economically reach, and feeding normalised data upward into it. It does not replace safety-rated control logic.

2. The four-step chain

Most competitors stop after the first step, which is why an alarm reaches a phone and nothing else happens.

01 · MONITOR

Connect what exists

Mixed-vendor equipment on one platform, over cellular or the site's own network.

02 · PREDICT

Read the warning

Per-asset baselines catch degradation — rising draw, longer cycles, drifting chemistry — before the failure it becomes.

03 · DISPATCH

Land the work

An alarm becomes a work order in the servicing partner's own system, routed to whoever is accountable.

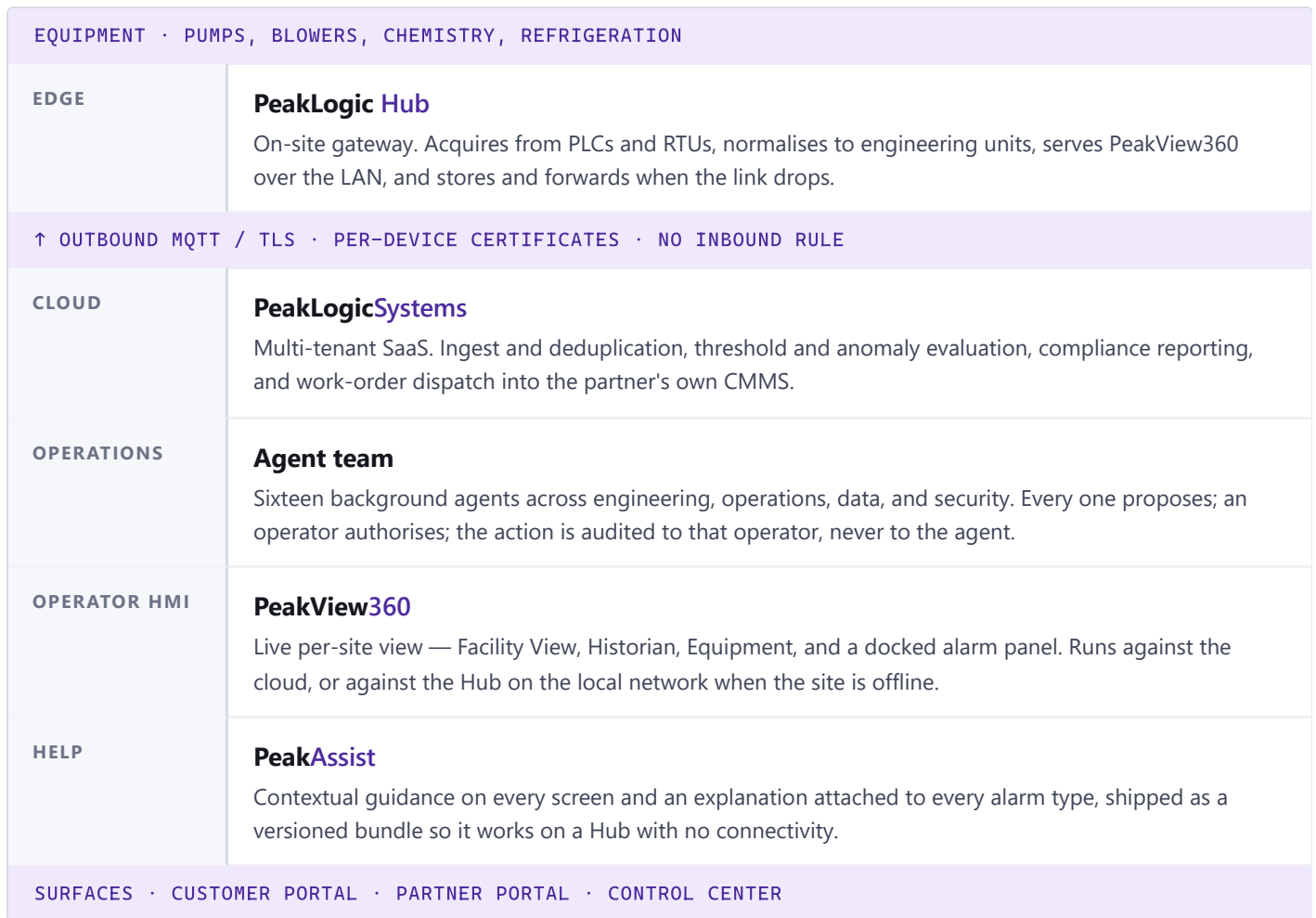
04 · PROVE

Keep the evidence

The record a regulator, an insurer, or a disputing customer asks for, compiled as it happens.

3. Platform architecture

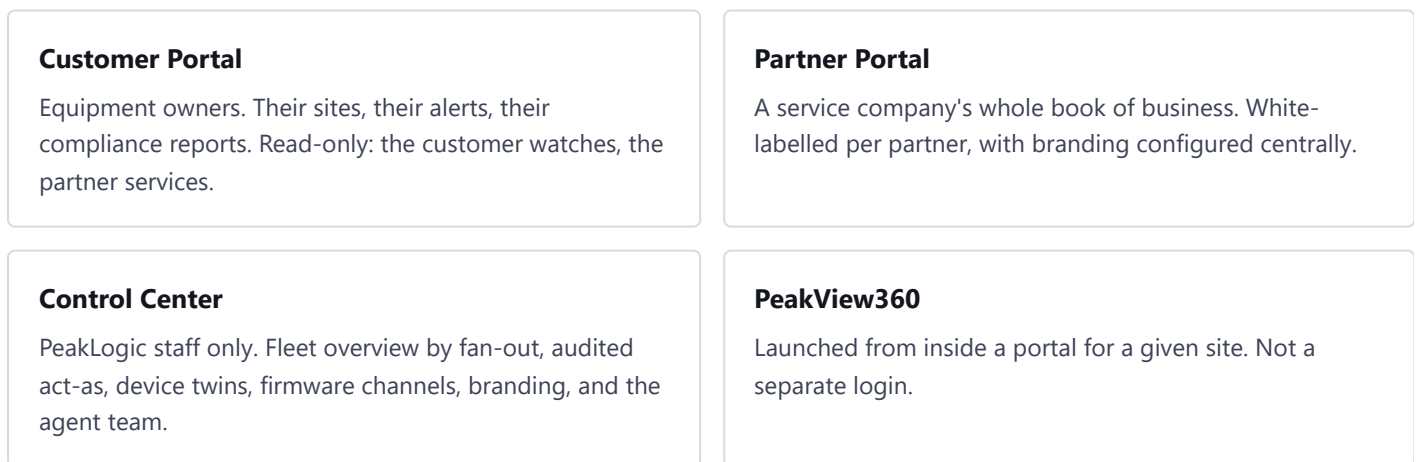
Five named layers. The two on the outside — equipment and the people who service it — are the only ones a customer thinks about.



Data moves upward only. No layer in this diagram issues a command downward to equipment — see §6.

Front-end surfaces

Three portals and one operator app — not one application with role-based menus.



4. Tenant isolation

Every tenant-scoped query runs inside a transaction that sets `app.current_tenant_id`, activating PostgreSQL row-level security. Partner and staff contexts have their own equivalents.

A site the platform holds no data for is **refused**, not approximated. The operator view renders nothing rather than falling back to another customer's facility — because a monitoring product that displays the nearest available plant is worse than one that displays none.

5. Product packs

Packaged by what the site is, because a lift station and a hotel pool need different thresholds, different reports, and different words.

PACK	CONTENTS
Lift Station	Control panel upgrade, pump alternation and level logic, fleet view, critical work orders
ATU Compliance	Panel or gateway, treatment monitoring, service reminders, compliance reports
Integrated Site	Every lift station and treatment unit on one property, one dashboard
WWTP Lite	Aeration, lift station, and disinfection monitoring for package plants
Pool	Chemistry, ORP, salt cell, filter pressure, pump speed, turnover
Cold Chain	Walk-in and rack temperature, compressor short-cycling, door events
Living Campus	Assisted living and healthcare: HVAC comfort, kitchen cold, generator exercise, DHW, leak detection

6. Capability by build state

Stated separately, because the credibility of the first list depends on not quietly merging it with the third.

BUILT AND TESTED

- **Telemetry ingest** with at-least-once deduplication, poison-message capture, and range sanitising. A bad PLC read never becomes a fabricated reading.
- **Threshold alerting** — per-category rule sets plus a config-driven Policy Engine with platform, tenant, site, and asset overrides. The compiled rules remain the permanent fail-safe: a safety monitor must never fail into silence.
- **Anomaly detection (Tier 1)** — per-device EWMA baselines and sigma scoring, capped at warning severity because it has no measured precision yet and so cannot dispatch a technician.
- **Device- and Hub-silence detection** — absence is a signal. Every threshold rule needs a value to test, so a dead sensor would otherwise be indistinguishable from a healthy one.
- **Connection-behaviour anomaly** — reporting cadence scored the same way a telemetry value is. An unusually short interval reads as flooding or replay; a long one is an early warning before full silence.

- **Hub agent integrity** — a reported agent version PeakLogic never published raises a tamper alert.
- **CMMS work-order lifecycle** — the dispatch funnel tracked on timestamps rather than a status enum, producing a real conversion metric and a service-visit record.
- **Compliance report generation** — DMR drafts with coverage-gap honesty: a parameter with no readings is reported as a gap, never interpolated. The operator remains filer of record.
- **PeakAssist** — authored content with governance tests that fail the build if a screen has no guide or an alarm type has no explanation.
- **Telemetry normalisation** — raw values to engineering units, with unmapped and rejected states that are never silently dropped.
- **Facility View** — projected isometric scene per site, with orthographic views and locate-by-sensor-type.

DESIGNED, NOT BUILT

- Predictive maintenance Tier 2 (failure-mode prediction) and prescriptive Tier 3 (LLM enrichment)
- Four of the sixteen agents, including the tenant-isolation prover — the highest-priority gap in the platform
- Compliance evidence packs, restore drills, secrets and certificate lifecycle

ROADMAPPED

Actuation. No code path issues a device command today and no permission grants one. Control is planned, because operators will want remote reset, setpoint change, and shutoff. The authorisation model, safety interlocks, and audit trail are being designed **before** any command path exists, and safety-critical trips will stay local to the equipment rather than depending on a cloud round trip.

Also roadmapped: Facility Builder authoring, site photography as a live-view backdrop. Cross-tenant benchmarking is explicitly **not** built and carries real legal caveats.

7. Deployment

Azure-native throughout: IoT Hub with DPS, Azure Functions on Flex Consumption, API Management, PostgreSQL Flexible Server with PostGIS, Microsoft Entra External ID across three isolated tenants, Key Vault, and Application Insights — all defined in Bicep.

Devices connect **outbound** over MQTT/TLS with per-device certificates. No inbound firewall rule, port forward, or dedicated VLAN is ever required on a customer network — which matters, because most target customers will not grant them.

No environment has been deployed. Infrastructure is compile-validated, and CI runs typecheck, unit tests, integration tests against real PostgreSQL, dependency audit, and best-practice checks on every push. Everything above describes code that exists and is tested — not code that is running in production.

8. What PeakLogic will not do

- Replace a permitted, safety-rated PLC without an equipment-maker partner
- Submit filings to a regulator, or assume regulatory responsibility
- Replace a licensed inspector
- Build a city-wide hydraulic model
- Train models across tenants, or benchmark one customer against another

PeakLogic · Product Description · Artifact #40 v1.0 · Prepared 2 August 2026. Build states are accurate as of this date and are maintained in the repository alongside the code they describe. Confidential.